

use vitcampus

switched to db vitcampus

db.sales.insertMany([

```
    { "_id" : 1, "item" : "Americanos", "price" : 5, "size": "Short", "quantity" : 22, "date" :  
ISODate("2022-01-15T08:00:00Z") },
```

```
    { "_id" : 2, "item" : "Cappuccino", "price" : 6, "size": "Short", "quantity" : 12, "date" :  
ISODate("2022-01-16T09:00:00Z") },
```

```
    { "_id" : 3, "item" : "Lattes", "price" : 15, "size": "Grande", "quantity" : 25, "date" :  
ISODate("2022-01-16T09:05:00Z") },
```

```
    { "_id" : 4, "item" : "Mochas", "price" : 25, "size": "Tall", "quantity" : 11, "date" :  
ISODate("2022-02-17T08:00:00Z") },
```

```
    { "_id" : 5, "item" : "Americanos", "price" : 10, "size": "Grande", "quantity" : 12, "date" :  
ISODate("2022-02-18T21:06:00Z") },
```

```
    { "_id" : 6, "item" : "Cappuccino", "price" : 7, "size": "Tall", "quantity" : 20, "date" :  
ISODate("2022-02-20T10:07:00Z") },
```

```
    { "_id" : 7, "item" : "Lattes", "price" : 25, "size": "Tall", "quantity" : 30, "date" :  
ISODate("2022-02-21T10:08:00Z") },
```

```
    { "_id" : 8, "item" : "Americanos", "price" : 10, "size": "Grande", "quantity" : 21, "date" :  
ISODate("2022-02-22T14:09:00Z") },
```

```
    { "_id" : 9, "item" : "Cappuccino", "price" : 10, "size": "Grande", "quantity" : 17, "date" :  
ISODate("2022-02-23T14:09:00Z") },
```

```
    { "_id" : 10, "item" : "Americanos", "price" : 8, "size": "Tall", "quantity" : 15, "date" :  
ISODate("2022-02-25T14:09:00Z") }
```

]);

{

acknowledged: true,

insertedIds: {

'0': 1,

'1': 2,

```
'2': 3,
'3': 4,
'4': 5,
'5': 6,
'6': 7,
'7': 8,
'8': 9,
'9': 10
}
}
db.sales.aggregate([
{
  $group: {
    _id: '$item',
    averageQty: { $avg: '$quantity' },
  },
},
]);
{
  _id: 'Mochas',
  averageQty: 11
}
{
  _id: 'Lattes',
  averageQty: 27.5
}
```

```

{
  _id: 'Americanos',
  averageQty: 17.5
}
{
  _id: 'Cappuccino',
  averageQty: 16.333333333333332
}
db.sales.aggregate([
  {
    $group: {
      _id: "$item",
      averageQty: {
        $avg: {
          $multiply: ["$quantity", 2]
        }
      }
    }
  },
  {
    $sort: {
      averageQty: 1
    }
  }
])
{

```

```

    _id: 'Mochas',
    averageQty: 22
  }
  {
    _id: 'Cappuccino',
    averageQty: 32.666666666666664
  }
  {
    _id: 'Americanos',
    averageQty: 35
  }
  {
    _id: 'Lattes',
    averageQty: 55
  }
db.sales.aggregate([
  {
    $group: {
      _id: '$item',
      averageAmount: { $avg: { $multiply: ['$quantity', '$price'] } },
    },
  },
  { $sort: { averageAmount: 1 } },
])
{
  _id: 'Cappuccino',

```

```

    averageAmount: 127.33333333333333
  }
  {
    _id: 'Americanos',
    averageAmount: 140
  }
  {
    _id: 'Mochas',
    averageAmount: 275
  }
  {
    _id: 'Lattes',
    averageAmount: 562.5
  }
db.sales.aggregate([
  {
    $group: {
      _id: '$item',
      averageAmount: { $avg: { $multiply: ['$quantity', '$price'] } },
    },
  },
  { $match: {averageAmount:{$gt:150}} },
])
{
  _id: 'Lattes',
  averageAmount: 562.5
}

```

```
}  
  
{  
  _id: 'Mochas',  
  averageAmount: 275  
}  
  
db.sales.aggregate([  
  {  
    $group: {  
      _id: '$item',  
      itemCount: { $count: {} }  
    },  
  },  
  },  
])  
  
{  
  _id: 'Cappuccino',  
  itemCount: 3  
}  
  
{  
  _id: 'Americanos',  
  itemCount: 4  
}  
  
{  
  _id: 'Lattes',  
  itemCount: 2  
}  
  
{
```

```

    _id: 'Mochas',
    itemCount: 1
  }
db.sales.aggregate([
  {
    $group: {
      _id: '$item',
      itemCount: { $count: {} },
    },
  },
  {
    $match: { itemCount: { $gt: 2 } },
  },
]);
{
  _id: 'Americanos',
  itemCount: 4
}
{
  _id: 'Cappuccino',
  itemCount: 3
}
db.sales.aggregate([
  {
    $group: {
      _id: null,

```

```

        totalQty: { $sum: '$quantity' },
    },
},
]);
{
    _id: null,
    totalQty: 185
}
db.sales.aggregate([
{
    $group: {
        _id: null,
        totalQty: { $sum: '$quantity' },
    },
},
{ $project: { _id: 0 } },
]);
{
    totalQty: 185
}
db.sales.aggregate([
{
    $group: {
        _id: '$item',
        totalQty: { $sum: '$quantity' },
    },
},

```



```
    },
  ]);
{
  _id: 'Mochas',
  totalQty: 11
}
{
  _id: 'Lattes',
  totalQty: 55
}
{
  _id: 'Americanos',
  totalQty: 70
}
{
  _id: 'Cappuccino',
  totalQty: 49
}
db.sales.aggregate([
  {
    $group: {
      _id: '$item',
      totalQty: { $sum: '$quantity' },
    },
  },
  { $sort: { totalQty: -1 } },
```

```
]);  
  
{  
  _id: 'Americanos',  
  totalQty: 70  
}  
  
{  
  _id: 'Lattes',  
  totalQty: 55  
}  
  
{  
  _id: 'Cappuccino',  
  totalQty: 49  
}  
  
{  
  _id: 'Mochas',  
  totalQty: 11  
}  
db.sales.aggregate([  
  {  
    _id: 1,  
    item: 'Americanos',  
    price: 5,  
    size: 'Short',  
    quantity: 22,  
    date: 2022-01-15T08:00:00.000Z  
  }  
])
```

```
{  
  _id: 2,  
  item: 'Cappuccino',  
  price: 6,  
  size: 'Short',  
  quantity: 12,  
  date: 2022-01-16T09:00:00.000Z  
}
```

```
{  
  _id: 3,  
  item: 'Lattes',  
  price: 15,  
  size: 'Grande',  
  quantity: 25,  
  date: 2022-01-16T09:05:00.000Z  
}
```

```
{  
  _id: 4,  
  item: 'Mochas',  
  price: 25,  
  size: 'Tall',  
  quantity: 11,  
  date: 2022-02-17T08:00:00.000Z  
}
```

```
{  
  _id: 5,
```

```
item: 'Americanos',  
price: 10,  
size: 'Grande',  
quantity: 12,  
date: 2022-02-18T21:06:00.000Z  
}
```

```
{  
  _id: 6,  
  item: 'Cappuccino',  
  price: 7,  
  size: 'Tall',  
  quantity: 20,  
  date: 2022-02-20T10:07:00.000Z  
}
```

```
{  
  _id: 7,  
  item: 'Lattes',  
  price: 25,  
  size: 'Tall',  
  quantity: 30,  
  date: 2022-02-21T10:08:00.000Z  
}
```

```
{  
  _id: 8,  
  item: 'Americanos',  
  price: 10,
```

```
size: 'Grande',
quantity: 21,
date: 2022-02-22T14:09:00.000Z
}
{
  _id: 9,
  item: 'Cappuccino',
  price: 10,
  size: 'Grande',
  quantity: 17,
  date: 2022-02-23T14:09:00.000Z
}
{
  _id: 10,
  item: 'Americanos',
  price: 8,
  size: 'Tall',
  quantity: 15,
  date: 2022-02-25T14:09:00.000Z
}
db.sales.aggregate([
  {
    $group:{
      _id : '$size',
      totalAmount:{$sum:{$multiply:['$price','$quantity']}}
    }
  }
])
```

```
    },{$sort:{totalAmount:-1}}
  ]);
{
  _id: 'Tall',
  totalAmount: 1285
}
{
  _id: 'Grande',
  totalAmount: 875
}
{
  _id: 'Short',
  totalAmount: 182
}
db.sales.aggregate([
{
  $group: {
    _id: null,
    maxQty: { $max: '$quantity' },
  },
},
{
  $project: {
    _id: 0,
  },
},
],
```

```
]);  
  
{  
  maxQty: 30  
}  
  
db.sales.aggregate([  
  {  
    $group: {  
      _id: '$item',  
      maxQty: { $max: '$quantity' },  
    },  
  },  
]);  
  
{  
  _id: 'Lattes',  
  maxQty: 30  
}  
  
{  
  _id: 'Americanos',  
  maxQty: 22  
}  
  
{  
  _id: 'Cappuccino',  
  maxQty: 20  
}  
  
{  
  _id: 'Mochas',
```

```
    maxQty: 11
  }
  db.sales.aggregate([
    {
      $group: {
        _id: '$item',
        maxQty: { $max: { $multiply: ['$quantity', '$price'] } },
      },
    },
  ]);
  {
    _id: 'Cappuccino',
    maxQty: 170
  }
  {
    _id: 'Americanos',
    maxQty: 210
  }
  {
    _id: 'Lattes',
    maxQty: 750
  }
  {
    _id: 'Mochas',
    maxQty: 275
  }
```



```

db.sales.aggregate([
  {
    $group: {
      _id: null,
      maxQty: { $min: '$quantity' },
    },
  },
  {
    $project: {
      _id: 0,
    },
  },
]);
{
  maxQty: 11
}

```

```

db.sales.aggregate([
  {
    $group: {
      _id: '$item',
      minQty: { $min: '$quantity' },
    },
  },
]);
{
  _id: 'Mochas',

```

```
    minQty: 11
  }
  {
    _id: 'Cappuccino',
    minQty: 12
  }
  {
    _id: 'Americanos',
    minQty: 12
  }
  {
    _id: 'Lattes',
    minQty: 25
  }
db.sales.aggregate([
  {
    $group: {
      _id: '$item',
      maxQty: { $min: { $multiply: ['$quantity', '$price'] } },
    },
  },
]);
{
  _id: 'Mochas',
  maxQty: 275
}
```

```

{
  _id: 'Lattes',
  maxQty: 375
}

{
  _id: 'Americanos',
  maxQty: 110
}

{
  _id: 'Cappuccino',
  maxQty: 72
}

```

Aggregation in java :

```

73      System.out.println(doc.toJson());
74    }
75    // =====
76    // Aggregation operation

```

Document Inserted Successfully
 Student Records:
 {"_id": 101, "name": "vishwesh", "cgpa": 8.9}
 {"_id": 102, "name": "utkarsh", "cgpa": 8.5}
 Document Updated Successfully
 Updated Record:
 {"_id": 101, "name": "vishwesh", "cgpa": 9.1}
 {"_id": 102, "name": "utkarsh", "cgpa": 8.7}
 Updated Record:
 {"_id": "vishwesh", "averagecgpa": 9.1}
 {"_id": "utkarsh", "averagecgpa": 8.7}
 Successfully renamed
 Successfully unset
 Document Deleted Successfully

Code :

```
// =====  
  
    // Aggregation operation  
  
    // =====  
  
    AggregateIterable<Document> result=collection.aggregate(  
        Arrays.asList(  
            Aggregates.group("$name",Accumulators.avg(  
                "averagecgpa",  
                "$cgpa"  
            ))))  
    );  
  
    System.out.println("\nUpdated Record:");  
    for(Document doc:result)  
    {  
        System.out.println(doc.toJson());  
    }
```

Output :

```
{"_id": "vishwesh", "averagecgpa": 9.1}  
{"_id": "utkarsh", "averagecgpa": 8.7}
```